Degree-separated coding and congruence norms for a ten-integer consistency equation

Mathematical research prepared for ProveIt

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1 Degree-separated coding: a ten-integer equation

There is a completely fixed integer polynomial

$$F_{10}(t, g_0, h_0, k_0, l, w, x, y, z, m)$$

of exact total degree 3 671 340, evaluated by a division-free circuit of 522 binary arithmetic operations, for which ordinary mathematics formalizable in ZFC proves

$$\neg\exists(t, g_0, h_0, k_0, l, w, x, y, z, m) \in \mathbb{Z}^{10}\ (F_{10} = 0) \quad \Longleftrightarrow \quad \mathrm{Con(ZFC)}.$$

This combines the preceding degree-separated source architecture with the incoming congruence-norm finalizer. Neither ingredient alone is claimed as new. The contribution is their proved compatibility, including the constructive residue restrictions, and the smaller explicit combined circuit. The inherited compiler calibration and Sun arithmetic bridge remain stated mathematical dependencies. No consistency decision, global minimality claim, or new end-to-end proof-assistant certificate is asserted.

The structural change is to split the extraction convolution by homogeneous degree. A linear label may now collide with a quadratic label: their kernels are separate summands. This relaxes the finite isolation problem rather than merely finding a better placement for the previous problem. Elementary row operations, one shared positivity shift, borrow-tolerant target padding, and exact mask-population calibration complete the construction.

1.1 Fixed source and three homogeneous kernels

Use exactly the unshifted, strictly positive, 31-coordinate parent in `data/parent_quartic31.json`, with its complete original four-port binding $(\beta_{\mathrm{native}}, K_0, M_C, M_F)$. The four included compressed payloads have bit lengths $(1064921, 1070004, 1064921, 1064922)$; the last is `MF_source`, already shifted, and is not shifted again. `INSTANCE.json` is the active binding. No package-52 transport row or coefficient is spliced into this tuple. The pinned parent theorem is

$$\exists p \in \mathbb{Z}_{>0}^{31}\ r_1(p) = \cdots = r_{22}(p) = 0 \quad \Longleftrightarrow \quad \neg\,\mathrm{Con(ZFC)}.$$

Replace only r_{14} by $\rho_{14} = r_{14} + r_2$ and r_{15} by $\rho_{15} = r_{15} + r_3$; otherwise put $\rho_j = r_j$. The inverse row operations preserve the common zero set over every coefficient tuple. There are now 89 records: seven constant, 39 linear, and 43 quadratic.

In the parent coordinate order

$$(J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma,\\ N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_{\mathrm{aux}}, \Delta),$$

choose the positive, distinct weights

$$(w_1,\dots,w_{31}) = (1,7,61,36,16,85,20,67,43,11,34,13,96,80,29,50,\\ 2,109,110,112,116,91,53,55,27,101,92,9,107,37,60).$$

The row order and targets are

$$(2,22,8,15,9,11,3,17,13,18,12,20,4,21,19,1,6,10,16,14,7,5),\\ (28,85,286,384,593,602,642,731,906,966,1083,1212,1231,1393,\\ 1461,1471,1678,1719,1732,1880,2107,2129).$$

Let q_j denote the target assigned to row j, and write $\rho_j = \sum_\alpha a_{j,\alpha} p^\alpha$. For $d = |\alpha|$ put

$$e(\alpha) = \sum_i \alpha_i w_i, \qquad c_{j,\alpha} = \begin{cases} 2a_{j,\alpha}, & d = 0,1, \\ a_{j,\alpha} \prod_i \alpha_i!, & d = 2. \end{cases}$$

Define the fixed integer polynomials

$$D_d(Z) = \sum_{j,\,|\alpha|=d} c_{j,\alpha} Z^{q_j - e(\alpha)}, \qquad U_p(Z) = \sum_i p_i Z^{w_i}, \qquad G_p(Z) = D_2(Z)U_p(Z)^2 + D_1(Z)U_p(Z) + D_0(Z).$$

All reversed exponents are nonnegative. The exact finite certificate checks, for every used degree-d monomial α and every degree-d monomial γ in 31 variables,

$$q_j - e(\alpha) + e(\gamma) = q_k \quad \Longleftrightarrow \quad j = k \text{ and } \alpha = \gamma.$$

There are respectively $1, 31, 496$ expansion monomials in degrees $0, 1, 2$, and 22,544 source/expansion interactions. There is no requirement to separate a degree-one label from a degree-two label. Multinomial expansion and the displayed coefficient weights therefore give the universal identities

$$[Z^{q_j}] G_p(Z) = 2\rho_j(p) \quad (1 \le j \le 22). \tag{1}$$

The independent checker compares the entire extracted polynomials in all 31 coordinates and four ports, not just evaluations at test points.

Write $M = 116$, $Q = 2129$. The exact kernel degrees and convolution height are

$$\deg D_0 = 1719, \qquad \deg D_1 = 2027, \qquad \deg D_2 = 1929, \qquad H = \max_d(\deg D_d + dM) = 2161.$$

The leading term of D_2 is exactly $+Z^{1929}$, arising from EX in row five. Every kernel separately has distinct exponents. Replacing each port monomial in a coefficient by its strict bit bound and taking absolute integer coefficients gives a maximum majorant of bit length 2134927. Consequently, with

$$C_0 = 2^{2134927}, \qquad \kappa = 2134940, \qquad \beta_S = 2^\kappa,$$

every kernel coefficient has absolute value less than C_0, and $\beta_S > 4096 C_0$.

1.2 Source polynomials and pre-binary bounds

In ten unrestricted integer unknowns $(t, g_0, h_0, k_0, l, w, x, y, z, m)$ define first

$$f = 2t + 1, \quad b = 6t + 4, \quad g = bg_0, \quad h = bh_0, \quad k = b(3k_0 + 2).$$

The circuit variable names are $(f, g, h, k, l, w, x, y, z, m)$; its first four names denote the free coordinates (t, g_0, h_0, k_0), not these computed registers. Then define

$$b = 1 + 3f, \quad B = \beta_S b^2, \quad \ell = \sum_i B^{w_i}, \quad U_c = \ell + g,$$

$$G_c = D_2(B)U_c^2 + D_1(B)U_c + D_0(B), \quad A_T = \sum_j B^{q_j},$$
$$N_0 = B^{117}, \quad N_1 = B^{2162}, \quad N = N_0N_1 = B^{2279},$$
$$M_0 = N_0 - 1 - (b-1)\ell, \quad J_c = 2G_c + A_T,$$
$$S_c = g + J_cN_0, \quad T_c = M_0 + (B-2)A_TN_0,$$
$$X = N^2 + (S_c + T_c)N + N - 1 - S_c,$$
$$Y = \beta_S^{15}\, 2^{\kappa-21}B^{2402}b = 2^{2418\kappa-21}b^{4805}, \qquad \mathcal{C} = bB^{116} = 2^{116\kappa}b^{233}.$$

All powers have fixed nonnegative integer exponents. Every displayed register is an abbreviation, never an additional quantified unknown.

Assume $f \neq 0$, b is a square, and $0 \leq g < 2\mathcal{C}$. Then $b \equiv 1 \pmod 3$ implies $b \geq 4$, and $B > 4096C_0b^2$. Distinct positive weights give $\ell < 2B^{116}$, hence

$$B^{116} \leq U_c < 3bB^{116}.$$

In particular, without assuming b is a power of two,

$$D_2(B) > \tfrac{1}{2}B^{1929}, \quad |D_1(B)| < 2C_0B^{2027}, \quad |D_0(B)| < 2C_0B^{1719}.$$

After division by $B^{1929}U_c^2$, the two possible negative terms are bounded by $2C_0/B^{18} < 1/8$ and $2C_0/B^{442} < 1/8$. Thus $G_c > \frac{1}{4}B^{1929}U_c^2 > 0$. This is the necessary signed leading-layer argument; positivity is not inferred from positive inputs alone.

The kernel degree bounds also give

$$|G_c| < 2C_0(1 + 3b + 9b^2)B^H \leq 26C_0b^2B^H, \qquad 2G_c < N_1/64.$$

Since $Q = 2129 < H$, both A_T and $(B-2)A_T$ are less than $N_1/64$. It follows that

$$0 < J_c < N_1/32, \qquad J_c + (B-2)A_T + 2 < N_1.$$

The digits of M_0 are $B-1$, except that the 31 designated cells have digit $B-b$. Therefore $0 \leq M_0 < N_0$, and $g < N_0$ as well. The preceding strict high-block bound proves

$$0 \leq S_c, \quad 0 \leq T_c, \qquad S_c + T_c < N. \tag{2}$$

Consequently $X \geq N^2 > 3b$ and $Y > \max(b, 256)$. Because κ is even, $B \equiv N \equiv 1 \pmod 3$, $M_0 \equiv 0$, and $T_c \equiv -22$; hence $X \equiv 1 + T_c \equiv 0 \pmod 3$. These are all established before binary-base recovery.

1.3 Borrow-tolerant extraction and exact mask population

Suppose now that b is a power of two. So are B, N_0, N. The low mask condition $g \,\&\, M_0 = 0$ is exactly

$$g = \sum_i n_iB^{w_i}, \quad 0 \leq n_i < b.$$

For these digits put $p_i = n_i + 1$, so $U_c = \sum_i p_iB^{w_i}$ encodes positive parent coordinates without expanding a successor substitution in every residual. Write $h_s = [Z^s]G_p(Z)$ and let ϵ_s be the indicator of the 22 targets. All coefficients, including unsupported collisions, satisfy

$$|h_s| < C_0\{(\sum_i p_i)^2 + \sum_i p_i + 1\} \leq 993C_0b^2 < B/4.$$

The signed expansion of J_c is $\sum_{s=0}^{H}(2h_s + \epsilon_s)B^s$. Euclidean digit normalization has incoming carries only in $\{-1, 0\}$:

$$u_0 = 0, \quad u_{s+1} = \left\lfloor \frac{2h_s + \epsilon_s + u_s}{B} \right\rfloor, \quad d_s = 2h_s + \epsilon_s + u_s - Bu_{s+1}.$$

At a marked cell the mask $B-2$ admits precisely digits $0,1$. If $h_s=0$, the unnormalized digit $1+u_s$ passes. If $h_s\ge 1$, it lies in $[2,B)$ and fails. If $h_s\le -1$, it lies strictly between $-B+1$ and zero, and its normalized digit is at least two, so it fails. Thus incoming borrows do not affect the zero test. Positivity of J_c was proved independently, ensuring these are its ordinary digits. Combining the disjoint low and high blocks with (1) proves

$$S_c \,\&\, T_c = 0 \quad\Longleftrightarrow\quad g=\sum_i n_i B^{w_i},\ 0\le n_i<b,\quad r_1(n+1)=\cdots=r_{22}(n+1)=0.$$

For $N=2^u$ and $S,T\ge 0$ with $S+T<N$, let $X_*=N^2+(S+T)N+N-1-S$. Its disjoint binary blocks give

$$s_2(X_*)=1+u+s_2(T)-\{s_2(S)+s_2(T)-s_2(S+T)\}.$$

The braces count carries with multiplicity, and vanish precisely when $S\,\&\,T=0$. Legendre's formula therefore implies

$$2^{1+u+s_2(T)}\mid\binom{2X_*}{X_*}\quad\Longleftrightarrow\quad S\,\&\,T=0.$$

The stronger range (2) is essential. Writing $b=2^k$, $B=2^v$, the actual mask population is

$$s_2(T_c)=(117+22)v-31k-22,\qquad u=2279v.$$

Hence $1+u+s_2(T_c)=2418v-31k-21=\log_2 Y$, exactly explaining the fixed polynomial threshold used above.

Conversely, a positive parent zero is not the all-one vector: original row five equals one there. For arbitrarily large $b=4^s$ exceeding all its coordinates, encode $n_i=p_i-1$. Then $g\ge B>b$ and $g<\mathcal{C}$; the latter follows from $(b-1)\ell<bB^{116}$. All marked tests pass and $Y\mid\binom{2X}{X}$. This proves both directions of the entire source contract needed by the inherited finalizer, including its lower bound on forward witnesses and its pre-binary reverse bounds.

1.4 The Sun bridge registers

Set

$$L=lY,\quad U=2LX,\quad V=4gwY,\quad W_*=bw,$$
$$A=U(V+1),\quad B_*=2X+1,\quad C=B_*+(A-2)h,$$
$$P=U^2V,\quad K=X+1+k(P-2),\quad S=2A-5,\quad T=3W_*C-2(W_*^2-1),$$
$$Q=((P/2)^2-1)K^2+1,$$
$$\Delta=8\mathcal{C}^3gK^2-g^2(32(C-KL)^2\mathcal{C}^3+g^2K^2).$$

The notation $P/2$ is only an abbreviation for the division-free expression $(LX)UV$, as $U=2LX$; `Phalf` is exactly that expression in the circuit. Thus $4Q=(P^2-4)K^2+4$, the old auxiliary square radicand.

Put $\xi=zx$, and form Sun's modified Lucas certificate

$$D=(A^2-4)C^2+4,\qquad E=C^2D\xi,\qquad F=4(A^2-4)E^2+1,$$
$$G_*=2(A+2)(A-2)^2E^2-1,\qquad \Omega=16E,$$
$$H_*=C+(B_*-C)F+\Omega Fy,\qquad I_*=(G_*^2-1)H_*^2+1.$$

Finally set

$$\rho=T+Sy,\quad \Sigma=\xi S^5,\quad \Theta=(\rho^3-\rho S^2-S^3)(\rho^2+23S^2).$$

The external bridge input is precisely Sun (2021), Theorems 4.12–4.13, specialized to $p=P_{\text{theorem}}=2, Q_{\text{theorem}}=1$, with the modified Lucas certificate and residue interpolation of Sun (2026), Theorems 3.1 and 4.1 and Propositions 6.1–6.2. These give the following interface.

- From a source witness with $b \in 2\uparrow$, $0 < g < \mathcal{C}$, and the central-binomial divisibility, choose positive l, w and integers h, k so that Q is a square, $S \mid T$, $\Delta > 0$, $C = u_{2X+1}(A)$, and $K = u_{X+1}(P)$.
- Conversely, if b is square, $f^2 l^2 \Delta > 0$, $\xi \neq 0$, $4Q$ and DFI_* are squares, and $S \mid T$, the bridge recovers $b \in 2\uparrow$, $0 < g < 2\mathcal{C}$, and the binomial divisibility.
- On the forward branch, ξ can be made a multiple of any prescribed positive integer, and y can be assigned any residue modulo ξ while retaining the square certificate DFI_*.

The exact forward values of C, K are not an extra assumption: the proof of Sun's 2021 Theorem 4.12 defines $C = u_B(A, 1)$ immediately before equation (4.16), which defines $K = u_{QX+1}(U^P V, 1)$. The recurrence used here is $u_0(a) = 0, u_1(a) = 1, u_{n+1}(a) = a u_n(a) - u_{n-1}(a)$.

1.5 Congruences valid for every integer assignment

These identities do not assume positivity, a source solution, a Pell equation, or a zero of the final polynomial.

First, B, N_0, N are multiples of b^2. Therefore

$$X + 1 \equiv -g \equiv 0 \pmod{b}.$$

Also X is odd, because b, g, B, N_0, N are even. Since $b \equiv \beta_S \equiv 1 \pmod 3$, all powers of B equal 1 modulo 3. There are 22 targets. Modulo 3 one has $M_0 = 0$, $T_c = -22 = 2$, and hence

$$X \equiv 1 + (S_c + T_c) + 1 - 1 - S_c = T_c + 1 = 0 \pmod 3.$$

The displayed formula for Y has a factor b; hence $2b \mid U$ and $2b \mid A$. Consequently $b \mid P$ and $3 \mid P$. The normalization $k = b(3k_0 + 2)$ gives

$$K \equiv 0 \pmod{b}, \qquad K \equiv 1 + 2(-2) = 0 \pmod 3.$$

It follows that

$$Q \equiv 1 \pmod{b^2}, \qquad Q \equiv 1 \pmod 3, \qquad \gcd(b, Q) = 1.$$

In particular $Q \neq 0$, Q is odd, and for $r = bQ$ we have $r \equiv 1 \pmod 3$, $|r| \geq 2$.

Since $h = bh_0$,

$$C + 1 = 2(X + 1) + (A - 2)bh_0 \equiv 0 \pmod{2b}.$$

Consequently $4b \mid D$: both terms of $D = A^2C^2 - 4(C^2 - 1)$ are divisible by $4b$.

1.6 The equation and unconditional two-root separation

Define

$$r = bQ, \quad a = r + (3\Delta - 1)(rz)^2, \quad d = DFI_*, \quad Z = \Sigma m - \Theta.$$

The single equation is

$$\boxed{F_{10} = \left(Z^2 - \Sigma^2(a + d)\right)^2 - 4\Sigma^4 ad = 0.}$$

The crucial separation is small and unconditional. Namely,

$$\frac{a}{b} = Q + (3\Delta - 1)bQ^2 z^2 \quad \text{is odd}, \qquad \frac{d}{b} \quad \text{is even}.$$

Both displayed quotients are integers; they are not operations in the emitted polynomial. Since $b \neq 0$, this proves $a \neq d$ on every assignment. No magnitude comparison or large separating coefficient is required.

The following elementary lemma explains why this is enough. If integers $a \neq d$ and a rational number q satisfy

$$(q^2 - a - d)^2 = 4ad,$$

then $q \neq 0$, because at zero the difference is $(a-d)^2$. Define $u = (q^2+a-d)/(2q)$ and $v = q-u$. Expanding gives $u^2 = a, v^2 = d$. A rational square root of an integer is integral, so both a, d are integer squares and $q = u+v \in \mathbb{Z}$. This proof includes negative and zero candidate radicands without making any prior choice of real or complex square roots.

Now $S = 2A-5$ is odd and never zero. The polynomial $\pi(v) = (v^3-v-1)(v^2+23)$ has no rational root, so $\Theta = S^5\pi(\rho/S) \neq 0$. If $\Sigma = 0$, the final polynomial is $\Theta^4 \neq 0$. At any zero we therefore have $\Sigma \neq 0$, and in particular $\xi \neq 0$, $z \neq 0$. Apply the lemma with $q = Z/\Sigma$. It yields

$$a, d \in \mathrm{Sq}(\mathbb{Z}), \qquad \Sigma \mid \Theta.$$

The latter follows from $q = m - \Theta/\Sigma \in \mathbb{Z}$.

The standard quintic merger then gives $S \mid T$. For clarity: if for some prime p one had $v_p(T) < v_p(S)$, then $v_p(\rho) = v_p(T)$. The two factors of Θ have valuations $3v_p(T)$ and $2v_p(T)$, respectively. Thus $v_p(\Theta) = 5v_p(T) < 5v_p(S)$, contradicting $S^5 \mid \Theta$.

1.7 The combined square and positivity guard

Factor

$$a = r\{1 + (3\Delta - 1)rz^2\}.$$

The two factors are coprime. Neither is zero: $|r| \geq 2$, $z \neq 0$, and the second factor is 1 modulo r. Since their product is a square, each is a signed square and both have the same sign. But $r \equiv 1 \pmod 3$ excludes a negative square. Hence r and the bracket are positive squares.

Similarly $r = bQ$, $\gcd(b, Q) = 1$, and $b \equiv Q \equiv 1 \pmod 3$. Each factor is a signed square, and the same modulus excludes negative squares. Thus b, Q are positive squares. Since b is even and nonzero, $b \geq 4$, and therefore $r \geq 4$.

If $\Delta \leq 0$, then

$$1 + (3\Delta - 1)rz^2 \leq 1 - r \leq -3,$$

contradicting positivity of the bracket. Therefore $\Delta > 0$. Since $\mathcal{C} = bB^{116} > 0$, the formula for Δ gives $K \neq 0$ and $g > 0$. If $l = 0$, then $U = P = 0$ and $Q = 1 - K^2 \leq 0$, contradicting its positive-square value. Thus $l \neq 0$, and $f = 2t + 1 \neq 0$ was automatic. We have recovered $f^2l^2\Delta > 0$, exactly the old bridge hypothesis, without placing the factor f^2l^2 in the new guard.

Before applying the bridge, rewrite

$$\Delta = gK^2(8\mathcal{C}^3 - g^3) - 32g^2(C - KL)^2\mathcal{C}^3 > 0.$$

Since $g > 0$ and $K \neq 0$, this gives $g^3 < 8\mathcal{C}^3$, hence $0 < g < 2\mathcal{C}$. The pre-binary source bounds already proved now yield $X > 3b$ and $Y > \max(b, 256)$ independently of any bridge conclusion. All hypotheses of the reverse bridge now hold. It yields the binomial source contract, which decodes a positive zero of the pinned parent. This proves soundness for all signed integer assignments, including degenerate values of the individual bridge variables.

1.8 Completeness of the new congruence restrictions

Suppose the parent has a positive zero. Choose a sufficiently large $b = 4^s$, $s \geq 1$, in the cofinal source construction. Then $t = (b-4)/6 \in \mathbb{Z}$, and the encoded g is a multiple of b^2. In fact

$$X + 1 \equiv -g \equiv 0 \pmod{b^2}.$$

Apply the constructive bridge. It gives the exact values $C = u_{2X+1}(A)$, $K = u_{X+1}(P)$ and positive l, w, as stated above. We show its h, k already have the imposed forms.

The index $X + 1$ is even. The recurrence shows that $u_{2j}(P)$ is divisible by P. Since $2b \mid U$, we have $4b^2 \mid P$, and thus $b^2 \mid K$. Since $P \equiv 0 \pmod 4$, $v_2(P-2) = 1$. In

$$(P-2)k = K - X - 1,$$

the right side is divisible by b^2. As b is a power of two, $b^2/2 \mid k$, hence $b \mid k$. Also $P \equiv X \equiv 0 \pmod 3$ and $K \equiv 0 \pmod 3$, so $k \equiv 2 \pmod 3$. Because $b \equiv 1 \pmod 3$, the integer k/b is 2 modulo 3. It is exactly of the form $3k_0 + 2$.

For the other restriction, the recurrence gives $u_{2j+1}(A) \equiv (-1)^j \pmod{A^2}$. Since X is odd, $C + 1$ is divisible by A^2, hence by $4b^2$. Also $B_* + 1 = 2(X + 1)$ is divisible by $2b^2$. Thus $2b^2 \mid C - B_*$. The constructive A is a multiple of four, so $v_2(A - 2) = 1$. Therefore

$$h = (C - B_*)/(A - 2)$$

is divisible by b^2, in particular by b; write $h = bh_0$. All normalization restrictions thus preserve completeness, with no change to the constructive l, w, C, K.

These witnesses have $\Delta > 0$, and b, Q positive squares. Set $r = bQ$, also a positive square. The positive integer $\delta = 3\Delta - 1 \equiv 2 \pmod 3$ is nonsquare. Hence δr is a positive nonsquare. Pell's theorem gives integers $u, z > 0$ with

$$u^2 - \delta r z^2 = 1.$$

It follows that $a = ru^2$ is a square.

It remains to use the two free certificate variables without losing this chosen z. Since $S \mid T$, write $q = T/S$. First apply the residue interpolation theorem with prescribed divisor z to obtain its positive ξ. Then choose a residue class of y for which $\pi(q + y) \equiv 0 \pmod \xi$. The polynomial π has a root modulo every positive integer, so this residue exists. Interpolation supplies the square $d = DFI_*$, and $x = \xi/z$ is integral. Then $\Theta = S^5\pi(q + y)$ is divisible by $\Sigma = \xi S^5$. Choose

$$m = \Theta/\Sigma + \sqrt{a} + \sqrt{d} \in \mathbb{Z}.$$

The polynomial identity for F_{10} now gives $F_{10} = 0$.

For the local-root fact used here, the cubic $v^3 - v - 1$ has discriminant -23. At an odd prime other than 23, either it has a simple linear factor, or its irreducibility makes the discriminant a square modulo the prime, so $v^2 + 23$ has a simple root. At 23 the cubic has simple root 3. At 2, $-23 \equiv 1 \pmod 8$ is a square in $\mathbb{Z}_2$. Hensel lifting and the Chinese remainder theorem prove the assertion. The cubic has no rational root by the rational-root theorem, and neither does $v^2 + 23$. These are exactly the inherited quintic merger facts.

1.9 Circuit and exact degree certificate

The artifact `generated/sun10.json` is a complete division-free circuit, with ten unrestricted variables and 522 live binary arithmetic gates: 333 multiplications, 164 additions, and 25 subtractions. Multiplication by a numeral counts, reused gates count once, and construction of fixed coefficient leaves is excluded. All four payload integers are included, so this operation count is not a bit-complexity claim.

The decisive exact degree ledger is

R	X	Y	A	C	K	Q
$\deg R$	9116	4805	18730	18732	32654	130612
R	Δ	a	d	Σ	Z	F_{10}
$\deg R$	75623	336851	1648366	93652	93670	3671340

The top term of the final norm is $\Sigma^4 d^2$. The verifier computes the coefficient at the propagated upper degree after specializing all ten variables to one indeterminate. At degree 3671340, using the actual fixed payload coefficients, the residues modulo $101, 1009, 1000003$ are respectively $54, 951, 129927$. Each nonzero residue certifies the exact degree independently.

The supplied checks establish all 22,544 source-isolation interactions, 58 local circuit identities, the factored Lucas identity, the two-root norm, actual coefficient bindings, gate liveness, and the degree certificate. Additional finite borrow, population, source-size, congruence, and Lucas-recurrence tests supplement the universal arguments; they do not replace the imported arithmetic or compiler theorems. Both ordinary Python and optimized Python use explicit failures rather than disabled assertions.

Against our previous degree-separated ten-integer circuit, the degree falls from 8469528 to 3671340 and the operation count from 533 to 522. Against the incoming congruence-norm circuit, the degree falls from 9152284 to 3671340 and the count from 586 to 522. These comparisons do not claim global optimality.

Provenance and dependency boundary. The new tail is adapted from `con_zfc_congruence_norms_2026-10-23`, received in commit `7331974b696693a8eed2aa5f0f64f1c19d9569f0`; its complete proof is retained under `upstream/`. The source encoding is the split116 component of our preceding wave, preserved verbatim alongside it. The complete original four-port tuple is retained. Sun's 2021 Theorems 4.12–4.13 and 2026 Theorems 3.1, 4.1 and Propositions 6.1–6.2 remain explicit mathematical dependencies; the pinned source compiler theorem remains a dependency as well. This is not a new proof-assistant verification of the composite endpoint.